SPACE TECH · MULTI-LEVEL

Space Tech Program

Connect space systems to problems on Earth.

Foundation to CubeSat/CanSat-style industry project.

24+ weeks | 4 modules | 16 lessons

Explore the course →

Concepts, worked examples, assignments and a coherent course project.

WHO IT IS FOR · WHAT YOU WILL LEARN

Your next capability.

A deep-tech path through orbital foundations, embedded systems, spacecraft architecture, communications, avionics, AI/autonomy and mission delivery.

- Understand spacecraft systems engineering
- Build embedded and communication subsystems
- Use simulation and mission tools
- Complete a CubeSat/CanSat-style team project

Audience

- Engineering students
- Embedded developers
- Space-tech aspirants

Prerequisites

- Engineering or strong STEM foundation

FROM CONCEPT TO EVIDENCE

1 Mission · 2 Orbit · 3 Payload · 4 Ground · 5 Analyze · 6 Validate

One curriculum, three connected views

This guide summarizes the same modules and lessons shown on the course overview and in the learning workspace. The next pages explain the scope and practical evidence for every module.

DETAILED MODULE CURRICULUM · 1–2

Build the foundations.

The module names, lesson sequence and evidence below match the course learning workspace.

1 Space Systems Foundations

Design an educational Earth-observation mission that measures vegetation change. Start with the decision the data must support, not with a shopping list of satellite components.

1.1 Space mission lifecycle
A mission moves from a stakeholder need through concept studies, requirements, design, integration, verification, operations and disposal.

1.2 Orbital mechanics
An orbit determines where a spacecraft travels, how often it can observe a location and when it can contact a ground station.

1.3 Space environment
Vacuum, radiation, temperature cycling and debris create different design constraints from a terrestrial classroom.

1.4 Systems engineering
Systems engineering connects the mission objective to measurable requirements, interfaces and verification evidence.

You will produce: Mission brief, orbit calculation with units, and a requirements/verification matrix.

2 Spacecraft Subsystems

Turn the mission concept into compatible subsystem budgets. The payload is intermittent, the computer runs continuously and the radio transmits only during contact.

2.1 Power systems
Power is an instantaneous rate in watts; energy is accumulated demand in watt-hours.

2.2 Thermal and structures
Thermal design tracks where heat is generated and how it leaves the spacecraft.

2.3 ADCS
Attitude determination estimates orientation using sensors; attitude control changes orientation using actuators.

2.4 Subsystem interfaces
Subsystem interfaces define more than connector shape.

You will produce: Power and eclipse budgets, a thermal/load-path sketch, and an interface-control record.

DETAILED MODULE CURRICULUM · 3–4

Apply, integrate and demonstrate.

The module names, lesson sequence and evidence below match the course learning workspace.

3 Avionics and Communications

Create an educational telemetry chain from a simulated sensor to a ground dashboard. Every reading must remain interpretable when delayed, missing or corrupted.

3.1 Embedded systems
An embedded controller has limited memory, execution time and power.

3.2 Sensors and telemetry
Telemetry is a documented message, not just a number.

3.3 RF and ground stations
A radio link needs a budget: transmitter power, antenna gains, path loss and other losses determine received power.

3.4 Flight software concepts
Flight software is commonly organized around explicit operating modes and controlled transitions.

You will produce: Telemetry schema, packet-loss test log, ground display and recovery-state diagram.

4 Autonomy and Mission Project

Integrate the mission brief, subsystem budgets and telemetry prototype into a reviewed educational mission package. Prefer explainable behavior to unsupported autonomy claims.

4.1 AI and autonomy
Autonomy chooses an action without a person issuing every immediate command.

4.2 Simulation and testing
Simulation makes repeatable tests possible before hardware is available, but the model has limits.

4.3 Mission operations
Mission operations connects planning, command approval, execution and evidence review.

4.4 CubeSat/CanSat-style capstone
The capstone is a coherent mission concept plus a simulated or ground-based educational demonstrator.

You will produce: Mission demonstrator, traceable verification report, operations runbook and final review presentation.

PROJECTS AND WORKING METHODS

Mission concept and satellite-data application

Define a mission objective, reason about system trade-offs and demonstrate an application of satellite or geospatial data.

Mission concept and trade study

System and ground-segment diagram

Data application and validation notes

Tools and methods

- Python and Linux
- Embedded C/C++
- Simulation tools
- CAD, PCB and FPGA concepts

Tools are learning choices, not partner endorsements.

Inside module 2: a worked example

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Illustrative power budget:
Payload: 8 W x 0.25 = 2 W average
Computer: 2 W x 1.00 = 2 W average
Radio: 10 W x 0.10 = 1 W average
Average before losses = 5 W; simultaneous peak = 20 W
35-minute eclipse at 5 W needs 2.92 Wh at the loads
At 85% conversion efficiency and 80% usable capacity:
Required nominal energy >= 2.92/(0.85*0.80) = 4.29 Wh
Educational estimate only; aging, temperature and margins remain to be assessed.
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The 5 W average, 20 W peak and 4.29 Wh nominal-energy estimate answer different design questions. Preserve all three instead of replacing them with one “power requirement”.

How your work is reviewed

Show the artifact, explain your decisions, test the stated assumptions and identify limitations. Module assignments build toward the course project rather than replacing it with a single example.

DELIVERY, COMPLETION AND NEXT STEPS

Know what you are signing up for.

Delivery

Cohort-based learning · 24+ weeks

Confirm current schedule, cohort arrangements and enrollment terms with the academy.

Completion requirements

Complete the required coursework and module assignments, and pass the course assessment. A course completion certificate is available after those requirements are met.

What the learning workspace includes

Follow the modules

Expand the curriculum and open the exact lesson you need.

Apply the concepts

Read explanations, inspect examples and complete module practice.

Keep your evidence

Use resources, notes, discussion and assignments to support your learning.

Learning commitments, not outcome guarantees

Completion is not a placement, employment or performance guarantee. The course project is educational evidence; any production use requires appropriate review and validation.

Educational mission scope

The CubeSat/CanSat-style capstone is a simulated or ground-based educational demonstrator. Launch, flight qualification and radio-spectrum approvals are not included or implied.

Start with the curriculum. Continue with the first lesson.

Review your starting point, download this guide and explore the connected course experience.

Go to the first lesson →

Client-review edition. The static learning workspace saves demonstration progress locally; production enrollment, assessment, communications and certificate issuance are not connected.